

SAFETY DATA SHEET

Section 1: Identification

Product Name: Microneedle Patch

Recommended Use: Single-use, sterile medical device to assist transdermal delivery of actives. No drug is contained in the device.

Manufacturing Method: photolithography-derived forming

Restrictions on Use: Single use only. Sterile. Do not reuse or resterilize. Not for ingestion or ocular use.

Supplier: Spike Architectonics Co., Ltd. Address: 64/14–15, Moo 3, Phahonyothin Road, Khlong Nueng Subdistrict, Khlong Luang District, Pathum Thani 12120, Thailand. Tel: +66 629979649 Email: info@spike-arch.com

Section 2: Hazards Identification

GHS Classification: Not classified as hazardous. (Solid article; no expected hazardous chemical release under normal use.)

Label Elements: No signal word or pictograms required.

Other Hazards: Mechanical puncture risk if misused; minor skin irritation may occur in sensitive individuals. Device is not sterilized.

Section 3: Composition/Information on Ingredients

Microneedle Structure: PUA (medical-grade polyurethane acrylate) ~ 1%

Substrate/Backing: Cellulose fabric / nonwoven (medical-grade) ~ 99%

Actives: None (device only)

Residuals: PUA monomer residuals are very low and not expected to pose hazards in normal use.

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Section 4: First-Aid Measures

Inhalation: Not applicable for solid article; if dust arises during processing, move to fresh air.

Skin Contact: If irritation occurs, discontinue use and wash with water and mild soap. Seek medical advice if symptoms persist.

Eye Contact: Rinse cautiously with water for several minutes. Remove contact lenses if present and easy to do.

Ingestion: Rinse mouth; seek medical attention if symptoms occur. Do not induce vomiting unless directed by medical personnel.

Most Important Symptoms: Local irritation; mechanical puncture discomfort if misused.

Section 5: Fire-Fighting Measures

Suitable Extinguishing Media: Water spray, dry chemical, foam, CO₂

Specific Hazards: Combustion may produce CO/CO₂ and polymer decomposition products.

Protective Equipment: Standard protective gear and SCBA.

Section 6: Accidental Release Measures

Personal Precautions: Avoid creating dust during cutting/shredding.

Environmental Precautions: Prevent littering into waterways.

Methods for Cleanup: Collect as solid waste; dispose of according to local regulations.

Section 7: Handling and Storage

Handling: Single-use sterile device. Do not reuse or re-sterilize. Do not use if the sterile barrier pouch is damaged. Keep out of reach of children.

Storage: Store in original packaging in a cool, dry place (e.g., 15–25°C); avoid humidity, direct sunlight, and heat.

Incompatibilities: Strong organic solvents, strong oxidizers, excessive heat that may degrade PMMA.

Section 8: Exposure Controls/Personal Protection

Occupational Exposure Limits: Not applicable for the finished article.

Engineering Controls: Local exhaust when cutting/processing in bulk to minimize dust.

PPE: Gloves for bulk handling; safety glasses during processing. Not required for normal end-use per IFU.

Section 9: Physical and Chemical Properties

Appearance: Solid microneedle patch with PUA microneedles; color _____; essentially odorless

pH: Not applicable (solid)

Softening/Glass Transition: PMMA Tg ~ 65°C (general data)

Solubility: PUA insoluble in water; backing generally insoluble

Flammability: Not readily flammable; combustible at elevated temperatures

Other Data: No explosive properties expected

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Section 10: Stability and Reactivity

Stability: Stable under recommended conditions

Hazardous Reactions: None under normal use

Conditions to Avoid: Excess heat, moisture, aggressive organic solvents

Incompatible Materials: Strong oxidizers, strong acids/alkalis

Decomposition Products: CO/CO₂ and small organics from polymer breakdown

Section 11: Toxicological Information

Likely Routes of Exposure: Skin contact at application site

Acute/Chronic Effects: Low toxicity expected for medical-grade materials

Irritation: Not observed based on available biocompatibility testing

Sensitization: Non-sterilization.

Section 12: Ecological Information

Ecotoxicity: No significant hazards expected for a solid article

Persistence/Degradability: Polymer components may be persistent

Bioaccumulation: Not expected

Section 13: Disposal Considerations

Pre-use: Consider non-hazardous solid waste

Post-use on patients: Treat as infectious/biological waste per local regulations. Do not discard with general waste.

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Section 14: Transport Information

UN Number: N/A – Not regulated

Proper Shipping Name: Medical device, non-active, EO-sterilized – Not regulated

Hazard Class: N/A (Not regulated)

Packing Group: N/A

Special Precautions: Protect from moisture, heat, and crushing; maintain sterile package integrity

Section 15: Regulatory Information

Device Status: Non-active medical device, Non-sterilization.

Quality Management System: Manufacturing site certified to ISO 13485:2016 & EN ISO 13485:2016 (**Certificate No.:** MD 709901, **Certification Body:** BSI, **Expiry:** 26-08-2026.)

GHS/CLP: Not classified as a hazardous chemical for normal use

Medical Device Regulations: Thai FDA: Authorized/registered —device category/class: Partial 2, **license/registration number:** กท. สผ. 50/2565, **approval date:** 30-07-2024.

Section 16: Other Information

Preparation/Revision Date: 27-07-2025 **Version:** 1

Disclaimer: Information herein applies to the described device under normal use. Users must evaluate suitability for their specific applications.